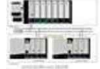


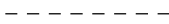
LEGEND (PROCESS MONITORING AND CONTROL SYSTEM):

EQUIPMENT LEGEND:

	MAIN PROCESS BACKBONE NETWORK SWITCH (FOR PMN AND PAN)
	PROCESS AUTOMATION NETWORK SWITCH (FOR SUB-RING AND SUB-SWITCHES)
	ROUTER
	PROGRAMMABLE LOGIC CONTROLLER (PLC) - HOT-STANDBY
	PLC - SINGLE CPU
	REMOTE INPUT/OUTPUT (RIO) PANEL (INCLUDING FOBOT FOR INCOMING AND OUTGOING F/O CABLE)
	MARSHALING PANEL
	SCADA I/O SERVER
	CYBER INTRUSION PREVENTION SYSTEM (IPS)
	SERVER
	NETWORK TIME SERVER
	SCADA CLIENT WORKSTATION
	OPERATOR INTERFACE PANEL (OIP) / HUMAN MACHINE INTERFACE (HMI)
	DATA DIODE
	SMALL FIREWALL
	PRINTER
	DOMAIN CONTROLLER
	CYBER INTRUSION DETECTION SYSTEM (IDS)
	FIBER OPTIC BREAKOUT TRAY
	MOTOR CONTROL CENTER (MCC)
	VARIABLE FREQUENCY DRIVE (VFD)
	SMART STARTER (DOL MOTOR STARTER)
	VALVE ACTUATOR
	F/O TO COPPER MEDIA CONVERTER
	FIELD INSTRUMENT
	FIELD ETHERNET SWITCH (MANAGED) WITH FIBRE AND COPPER PORTS
	PLC FOR COMPLEX MACHINE
	HOT-STANDBY PLC WITH I/O VENDOR PACKAGE OR DESIGN & BUILD PACKAGE
	MOTOR STARTER

	MCC DATA CONCENTRATOR
	TELCO/ISP INTERFACE EQUIPMENT
	CONTROLLER FOR AUXILIARY SYSTEM
	REMOTE TERMINAL UNIT

CABLE LEGEND:

	SINGLE-MODE F/O CABLES
	MULTI-MODE F/O CABLES
	COPPER CABLE
	COMMUNICATION WITH OTHERS
	CONTROL / SIGNAL CABLES

BOUNDARY LEGEND:

	CONTRACT BOUNDARY LINE
	LOCATION BOUNDARY
	FUTURE WORKS BOUNDARY LINE
	CONTRACT TERMINATION POINT
	VENDOR PACKAGE

ABBREVIATIONS :

PUBOC	PUBLIC UTILITIES BOARD OPERATION CENTRE
PMN	PROCESS MONITORING NETWORK
PAN	PROCESS AUTOMATION NETWORK

TENDER

T1	20/05/22	ISSUED FOR TENDER C4B	JL	GOT
T1	03/12/21	ISSUED FOR TENDER C3E	JL	GOT
T1	02/07/21	ISSUED FOR TENDER C3D2	LC	GOT
T1	18/12/20	ISSUED FOR TENDER C3B1	LC	GOT
T1	22/07/20	ISSUED FOR TENDER C2B	LC	GOT
T1	17/04/20	ISSUED FOR TENDER C3D1	LF	GOT
REV	DATE	DESCRIPTION	REVIEWED	AUTHORISED

CLIENT:



PROGRAMME MANAGER:



CONSULTANT:



PROJECT TITLE:

TUAS WATER RECLAMATION PLANT (TWRP)
DTSS PHASE 2

DRAWING TITLE

GENERAL LEGEND
AND ABBREVIATIONS
PROCESS

DRAFTED: PA	DESIGNED: SK	SHEET SIZE: A1
DRAFTING CHECK: RS	DESIGN CHECK: AT	SCALE: N/A
DATE: 20/05/22	REVIEWED: JL	PBS: 131101
CONTRACT: C4B	DRAWING NUMBER: PUB-DTSS2-TWRP	
	SHEET NUMBER: WSW00-I-0011	REVISION: T1

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